LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	24
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine vibration monitoring and indicating system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine vibration monitoring and indicating system.
5. Specific Objectives	15.1.9.1 Trainee will be able to explain Engine vibration monitoring and indicating system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of vibration indicator panel layout, safety precautions regarding hot engine components, instructor supervision during system explanation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of engine vibration?
(b) Why is monitoring engine vibration critical for aircraft safety?
(c) Can you identify the location of the vibration indicator on the cockpit panel?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of engine vibration generation • Types of vibration sensors and transducers • Signal conditioning and amplification circuits • Vibration indicator display mechanisms • Alarm and warning system integration	L & Ds	PPT, IB	15 min
(b)	• Interpretation of vibration indicator readings • Normal versus abnormal vibration thresholds • Troubleshooting common vibration faults • Maintenance procedures for vibration sensors • Impact of vibration on aircraft structural integrity	L & Ds	PPT, PDF	20 min
(c)	• System calibration and testing protocols • Record keeping for vibration data logs • Emergency procedures for excessive vibration • Coordination with ground engineering teams • Future trends in vibration monitoring technology	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of a piezoelectric vibration sensor.
(b) Describe the steps taken when the vibration indicator shows an abnormal reading.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of engine vibration monitoring discussed in this lesson apply to the magnetic components affecting compasses covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction, effects of magnetic components on compasses.	Refer to the master syllabus reference list	LP 25

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min